0580_w24_ms_43_Q12

Question	Answer	Marks	Partial Marks	Guidance
12(a)	$\sqrt{8.7^2+11.4^2-2\times8.7\times11.4\cos119}$	**M2**	**M1** for $8.7^2+11.4^2-2\times8.7\times11.4\cos119$ **A1** for 301.8...	
	17.372 to 17.373	**A1**		**M1** and 17.372 to 17.373 scores full marks
12(b)	13.[0] or 13.02 to 13.03	**4**	**M2** for $\sin E = \dfrac{17.37\sin 20}{10.9}$ or **M1** for $\dfrac{10.9}{\sin 20} = \dfrac{17.37}{\sin E}$ oe **M1** for $\angle ACE = 180 - 20 -$ *their* obtuse AEC oe	Allow more accurate 17.37 For M2 and M1 condone use of 17.4 (leads to $\sin E = 0.5450\ldots$, $E = 33.09\ldots$ ($E = 33.027...$ or $146.97...$)
12(c)	40.9 or 40.91 to 40.94	**3**	**M1** for a correct implicit trig statement for AD eg $\sin(\textit{their acute } ACE) = \dfrac{AD}{17.37}$ oe **M1** for a correct implicit statement for CD eg $\cos(\textit{their acute } ACE) = \dfrac{CD}{17.37}$ oe or $CD^2 = 17.37^2 - (\textit{their}AD)^2$ or for a correct statement for ED eg $\tan(180 - \textit{their}$ obtuse $\angle AEC) = \dfrac{\textit{their}AD}{ED}$	Allow more accurate 17.37 $\angle ACE = 13.0$ or 13.02 to 13.03 $AD = 3.91$ to 3.92 $CD = 16.92\ldots$ AD and CD may be in reverse order $ED = 6.0225\ldots$ Many ways of finding the perimeter. Finding AE may be common. $AE = 7.183$ to $7.184\ldots$